

Machine Learning Operations (MLOps)

Set of operational processes to manage ML models, their code, and the data they rely on to ensure reliable and trustworthy application of ML in real-world problems

Feature	DevOps	MLOps
Primary Focus	Software Application	Machine Learning lifecycle Management
Key Artifacts	Code Binaries	Code, Data, Models, Hyperparameters
Testing	Unit, Integration, End-to-End	Data validation, Model quality, Fairness testing
Versioning	Code	Code, Data, Model, Experiment versioning
Deployment Target	Executable application or microservice	Prediction service(API) or continuous training pipeline
Monitoring	Uptime, Logs, Latency, CPU/Memory usage	Uptime, Latency + Data Drift, Bias, Explain-ability, Concept Drift, Model Accuracy
Automation Flow	CI/CD (Continuous Integration/ Delivery)	CI/CD/CT (Continuous Integration/ Delivery/ Training)

Data Source

Types:

- DB backends
- API endpoints
- Messages/ Streams
- Batch files

Data Formats:

Structured (CSV, XML, JSON) vs unstructured (text, binary)

Data inside software: All software is backed by RDBMS or NoSQL.

Types of data in a DB:

- Facts/ Events – Describe a real-world event or business process
- Dimension/ Masters – provide favour to that description that allow interpretation in the real world.

Data Definitions and Changes

- For facts, append
- For dimensions, CRUD Create, Read, Update, Delete

CRUD Implementation types:

- In-place updates: Overwrite old data (no history kept).
- Appends: Add new rows for updates (history kept, but harder to query current state).
- Slowly Changing Dimension (SCD) Type 2: Use effective dates to track historical changes for an entity while keeping one row marked as "current." This is the industry standard for maintaining accurate historical data for analytics and ML training.

Formats & Versioning: Complex non-relational data (JSON/XML) must be "shredded" or "flattened" into relational tables (CSV/SQL) for effective ML training. Crucially, always log both

the raw source and the final prepared data to ensure you can debug your model if something goes wrong.

Data Integration/ Ingestion:

- Publish service and allow pulls – common. Technical Service, data formatted in specific way and Functional Service, data organised into logical, real world business concepts.
- Call APIs and push -rare, mainly for real time
- Do nothing, or even make it hard!

Google Cloud